



X(6th Sm.)-Computer Sc.-HI/(DSE-A-4)/CBCS

2022

COMPUTER SCIENCE — HONOURS

Paper : DSE-A-4

(Multimedia and its Applications)

Full Marks : 50

The figures in the margin indicate full marks.

Candidates are required to give their answers in their own words as far as practicable.

Answer *question no. 1* and *any four* questions from the rest.

1. Answer *any five* questions : 2×5
 - (a) Name the colour model used most in printing trade and the colour model that represents colour as an angle from 0 to 360 degrees on the colour wheel.
 - (b) List the three commonly used sampling rates often used in multimedia for digital audio.
 - (c) Name the most widely used tool for creating animation nowadays. Why is it so popularly used?
 - (d) How is lossy-to-lossy trans-coding different from lossless-to-lossy trans-coding?
 - (e) What is SCSI?
 - (f) "Transcoding is used extensively by home theatre PC". Why?
 - (g) How is MPEG-4 different from MPEG-2?
 - (h) How is overscan different from underscan in video displaying?
2. (a) Explain with examples, any four principles of animation needed for effective animation. (2×4)+2
 - (b) What is Chroma keys?
3. (a) Explain in detail, the four basic stages of a multimedia project. (2×4)+2
 - (b) Name any two features that are a must for a good image editing tool.
4. (a) "AVI video format is a good choice for home video recordings, whereas WEBM video format is a good choice for online videos".— Give reasons.
 - (b) If a multimedia project is built on a Windows platform and then played on a Macintosh platform, will there be any difference? How does font substitution and font mapping help one in such a situation?
 - (c) Explain with an example, the usefulness of hypertext when going through a text document with hypertexts and one without any hypertexts. 3+(1+2)+4

Please Turn Over

5. (a) What is 3D modelling? Describe different ways of 3D modelling.
- (b) What is additive color model and subtractive color model? Give examples and state why we use additive or subtractive model respectively in the examples stated above. (2+2)+(2+4)
6. (a) List down the steps that one needs to follow to bring an audio recording into a multimedia project, either in Windows or in Macintosh.
- (b) For music files, 128 Kbps sampled MP3 files are good to go, whereas human voice can be sampled at 64 Kbps.— Give reason. 8+2
7. (a) List down in detail, the different variety of input devices (other than traditional ones like mouse or keyboard), that can be used for the development and delivery of a good multimedia project. 5+5
- (b) List down the key features that a good video conferencing software should possess.
8. (a) "The color white is a noisy mixture of all the colors in visible spectrum". Cite four examples that justify this statement.
- (b) Describe the capabilities and limitations of bitmap, vector and 3-D image representations. 4+(3+3)